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Practical 2

Aim : Create a Blockchain using Python

Theory

Blockchain is a distributed digital ledger in which data is stored in blocks connected using cryptographic hashes. Each block contains transactions, timestamp, nonce, previous hash, and its own hash. Mining is the process of finding a valid nonce using Proof of Work. In this experiment, the hash must contain three leading zeros (000). Blockchain validity is checked by verifying the hashes and links between consecutive blocks.

Tasks Performed

1. Corrected errors in the given Blockchain program.
2. Added the create_transactions() method.
3. Modified mining to occur only when transactions are available.
4. Removed transactions from the pending list after mining.
5. Modified proof_of_work() to find the golden nonce.
6. Implemented the condition of 000 leading zeros in the hash.
7. Tested the Blockchain using Flask and Postman.
8. Displayed the Blockchain and checked its validity.

Program and Output

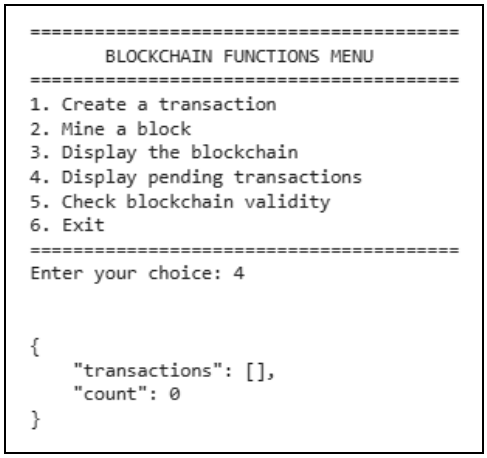
```
========================================
       BLOCKCHAIN FUNCTIONS MENU
========================================
1. Create a transaction
2. Mine a block
3. Display the blockchain
4. Display pending transactions
5. Check blockchain validity
6. Exit
========================================
Enter your choice: 1
Enter sender: ABC
Enter receiver: XYZ
Enter amount: 1000

Transaction created successfully!
{'sender': 'ABC', 'receiver': 'XYZ', 'amount': 1000.0}
```

```
========================================
       BLOCKCHAIN FUNCTIONS MENU
========================================
1. Create a transaction
2. Mine a block
3. Display the blockchain
4. Display pending transactions
5. Check blockchain validity
6. Exit
========================================
Enter your choice: 4

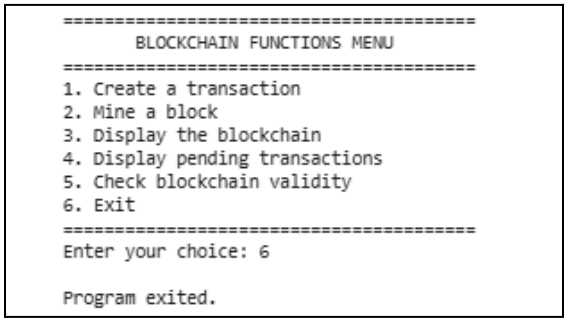

{
    "transactions": [
        {
            "sender": "ABC",
            "receiver": "XYZ",
            "amount": 1000.0
        }
    ],
    "count": 1
}
```











Conclusion

The Blockchain was successfully implemented using Python. The experiment demonstrated transactions, mining, Proof of Work, golden nonce, hashing, and Blockchain validation. It helped in understanding the basic working and security of Blockchain technology.